Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Jayden Lugo	jaylugo@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Rowers train on the ERG (indoor rowing machine) constantly, but it's hard to see progress across weeks of workouts. The machine only shows one workout, and it's not easy to compare times, splits, and watts between different tests. My goal is a simple way to log erg workouts, convert them into the numbers rowers actually use (500m split and watts), and track personal records.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

The ErgData app syncs workouts directly from the erg onto the app to keep a full history, but they're built around the hardware and have a lot of features a beginner wouldn't need. There's also other general fitness apps like Strava, but they aren't focused on rowing-specific numbers like 2k goals.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

My site will be an erg training log. A rower enters their workout, (distance, time, stroke rate, and workout type), and the website calculates the 500m split and average watts. It will store the workout in the history, and flag new personal records. Users can set a 2k goal time, and see how far their best 2k is from that time.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program will take data from text boxes, then stores each workout as a Workout dataclass instance . It will take that data and calculate splits, watts, totals, and personal records on the page

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Workout distance in meters Workout time in minutes + seconds Average stroke rate Workout type (steady state, interval, or test) Goal 2k time name

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

Convert time to total seconds Calculate 500m split Calculate average watts Compare new workout against stored to decide if it is a personal record Filter different workout types

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

A page showing the inputted workout showing split watts, and if a pr, “new pr!”
History page showing all previous workouts
Stats page with total meters, avg split, and best 2k

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Workout : one logged erg piece
Rower : the current session with the rower’s name, workouts (list[Workout]) and goal 2k

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

For Workout, distance in meters, time in seconds, average stroke rate, workout type
Ignore heart rate, drag factor, date/time, rest time, calories

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Passing data between routes correctly with drafter so a workout is properly saved.
Many different formulas for calculating every different statistic

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow

learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

If I use an llm, I will make sure to have it explain concepts and help debug, not write the project for me. I'll write my code one part at a time, to make sure I don't get stuck and larger problems don't appear.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

I want to build something that will actually be useful for my own training

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple app that helps rowers log and track their erg workouts. What are 3 common problems rowers face when tracking their erg training, and what are 3 simple features a basic tool should have?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

Hard to compare across workouts, scores get lost as they get tracked in various places, and lack a clear goal. It should have easy workout logging, workout history, and goal tracking with automatic conversions. It also suggested being able to connect directly to the erg.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it

is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Syncing workouts automatically from the monitor is way too complicated for an intro programming project. It would mean connecting to the erg over Bluetooth and drafter would not be able to handle it.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

The conversation helped me realize what parts of the project are actually feasible, and what is actually necessary in an app like this.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

The course chapters on dataclasses, routes, and drafter
Earlier assignments to use for reference
TA / instructor office hours

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Read the full error
Try and figure out what line is causing the error
Run more tests
Office hours

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I want to build a simple app that helps rowers log and track their erg workouts. What are 3 common problems rowers face when tracking their erg training, and what are 3 simple features a basic tool should have?

Here's a breakdown.

3 common problems rowers face tracking erg training

1. **Numbers are hard to compare across workouts.** A 2k test, 4×1000m intervals, and a 30-minute steady state all show different totals, so it's hard to tell whether you're actually getting fitter. The 500m split is the common language, but working it out by hand from time and distance is tedious.
2. **Scores get lost.** Many rowers jot results in their phone notes, on a whiteboard at practice, or not at all. Without one place for them, it's hard to know your personal records or spot trends.
3. **No clear link between training and goals.** Rowers often have a target 2k time but don't know what split that requires or how far off they currently are.

3 simple features a basic tool should have

1. **Workout logging with automatic split and watts.** Enter distance, time, stroke rate, and workout type, and the app calculates the 500m split and average watts.
2. **Workout history with personal records.** A list of all past workouts, filterable by type, with the app flagging a new best when you beat your old 2k.
3. **Goal tracking.** Set a goal 2k time and see the split needed to hit it and the gap between your current best and the goal.

Bigger features you could add later

- Progress graphs showing your split over weeks
- Syncing workouts automatically from the erg's PM5 monitor
- Auto-generated training plans that build a weekly schedule toward a test date
- Team features so a coach can see every athlete's scores

Existing tools

- **Concept2 Logbook and the ErgData app:** the official tools. They sync from the erg monitor and keep a full history, but they're tied to the hardware and include a lot of features.
- **Online pace calculators**, such as Concept2's: convert time and distance into split and watts, but don't save anything.
- **ErgZone and Strava:** workout tracking with more of a general fitness or social focus.
- **Spreadsheets:** many teams log scores in Google Sheets, which means doing the math by hand.

For a simple app, the three core features are a solid scope. The bigger additions are where things get complicated quickly.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.